

Use descriptive names for sets. Once configured, use the `mongo` shell to add hosts to the replica set.

See also:

Replica set reconfiguration.

To enable authentication for the *replica set*, add the following `keyFile` option:

In YAML format

```
security:
  keyFile: /srv/mongodb/keyfile
```

Or, if using the [older configuration file format](#)⁴⁸:

```
keyFile = /srv/mongodb/keyfile
```

Setting `keyFile` enables authentication and specifies a key file for the replica set member use to when authenticating to each other. The content of the key file is arbitrary, but must be the same on all members of the *replica set* and `mongos` instances that connect to the set. The keyfile must be less than one kilobyte in size and may only contain characters in the base64 set and the file must not have group or “world” permissions on UNIX systems.

See also:

The [Replica Set Security](#) (page 323) section for information on configuring authentication with replica sets.

The [Replication](#) (page 559) document for more information on replication in MongoDB and replica set configuration in general.

Sharding Configuration Sharding requires a number of `mongod` instances with different configurations. The config servers store the cluster’s metadata, while the cluster distributes data among one or more shard servers.

Note: *Config servers are not replica sets.*

To set up one or three “config server” instances as [normal](#) (page 193) `mongod` instances, and then add the following configuration option:

In YAML format

```
sharding:
  clusterRole: configsvr
net:
  bindIp: 10.8.0.12
  port: 27001
```

Or, if using the [older configuration file format](#)⁴⁹:

```
configsvr = true

bind_ip = 10.8.0.12
port = 27001
```

This creates a config server running on the private IP address `10.8.0.12` on port `27001`. Make sure that there are no port conflicts, and that your config server is accessible from all of your `mongos` and `mongod` instances.

To set up shards, configure two or more `mongod` instance using your [base configuration](#) (page 193), with the `shardsvr` value for the `sharding.clusterRole` setting:

⁴⁸<http://docs.mongodb.org/v2.4/reference/configuration-options>

⁴⁹<http://docs.mongodb.org/v2.4/reference/configuration-options>